
Congestion Structure and Exceedance Bounds for Locational Marginal Emissions

Cameron Khanpour
Georgia Institute of Technology
khanpour@gatech.edu

Samuel Talkington
University of Michigan
talks@umich.edu

Daniel K. Molzahn
Georgia Institute of Technology
molzahn@gatech.edu

Abstract

Locational marginal emissions (LMEs) give the sensitivity of total operating carbon emissions to nodal power demand. We show that this vector with n entries has a much smaller intrinsic dimension under DC optimal power flow. Within a fixed active constraint set, the LME vector lies in the span of the uniform vector and the power transfer distribution factor rows of the binding lines. Its rank r is therefore at most one more than the number of binding/congested lines. This structure makes r independent scalar observations necessary and sufficient for exact recovery. Across ten systems with nonzero operating emissions, from 14 to 1,354 buses, r ranges from 2 to 15. For instance, on a 300 bus system, 24 dispatch simulations recover all 300 LMEs. We also derive an emissions exceedance bound under uncertain demand. The bound separates variation while the nominal active set remains unchanged, the probability of an active set change, and estimation error. Numerical results show that its usable forecast error range depends on local active set geometry.

1 Introduction

Locational marginal emissions (LMEs) measure the change in total operating emissions caused by one additional MWh of energy consumed at each bus during a dispatch interval. Prior work uses these sensitivities to guide flexible demand, including data center load shifting outside the system operator's dispatch [Lindberg et al., 2021, 2022]. Other formulations place carbon objectives inside dispatch [Chen et al., 2025, Sun et al., 2023, Gu et al., 2022]. However, changing the set of active constraints changes the LMEs, and deterministic validity ranges can be small [Nilges et al., 2025].

We make the following two contributions. First, we derive an intrinsic basis for the LME vector, prove that its rank r is at most one more than the number of binding line flow constraints, and show that r independent observations are necessary and sufficient for recovery. Second, we combine a forecast error distribution with the set of demand changes that preserve the nominal active set, which we call the critical region. The resulting probability bound separates emissions variation inside the region, region exit, and estimation error. Numerical experiments test both results across networks up to 1,354 buses.

Existing methods for computing LMEs use direct differentiation [Fuentes Valenzuela et al., 2024], LMP mappings within critical regions [He et al., 2024], or finite differences at *every* bus. We focus on LME recovery when an external or legacy dispatch simulator can be evaluated but its equations or solver derivatives cannot be accessed.

2 Congestion Basis and Simulation Count

Let $d \in \mathbb{R}^n$ denote net demand in MW and $g \in \mathbb{R}^{n_g}$ generation in MW for a one hour dispatch interval, Γ the generator incidence matrix, and H the power transfer distribution factor (PTDF)

matrix, whose entry H_{ji} gives the flow on line j from an injection at bus i . We study the strictly convex DC dispatch

$$g^*(d) = \arg \min_g \left\{ \frac{1}{2} g^\top Q g + a^\top g \mid \mathbf{1}^\top (\Gamma g - d) = 0, |H(\Gamma g - d)| \leq \bar{f}, \underline{g} \leq g \leq \bar{g} \right\}, \quad (1)$$

where $Q \succ 0$ is the quadratic cost matrix. Total hourly operating emissions are $\mathcal{E}(d) \triangleq c^\top g^*(d)$ for generator emission rates c . Fix a nominal demand d_0 with a regular active set. This means that the active constraint rows are linearly independent and their inequality multipliers are positive. The demands preserving this active set form its critical region, within which $g^*(d)$ is affine. Let $\mathcal{J}$ index the binding lines and let $s_j \in \{-1, 1\}$ denote the binding direction. The active equalities are

$$\mathbf{1}^\top \Gamma g = \mathbf{1}^\top d, \quad s_j H_j \Gamma g = \bar{f}_j + s_j H_j d \quad (j \in \mathcal{J}), \quad g_k = \underline{g}_k \text{ or } g_k = \bar{g}_k \quad (2)$$

for each binding generator limit k . After dependent rows are removed, let K stack the coefficients multiplying g in Equation (2) and write the right side as $b + Dd$. The rows of D are $\mathbf{1}^\top$, the signed PTDF rows $s_j H_j$, and zeros for generator limits. Differentiating the first-order optimality conditions gives

$$G \triangleq \partial g^* / \partial d|_{d_0} = Q^{-1} K^\top (K Q^{-1} K^\top)^{-1} D, \quad \ell \triangleq G^\top c, \quad (3)$$

where ℓ_i is the carbon emissions change per additional MW of demand sustained for the one hour interval, with units of tonnes CO2/MWh. Equivalently,

$$\ell = D^\top (K Q^{-1} K^\top)^{-1} K Q^{-1} c. \quad (4)$$

Equation (4) shows that the cost and generator data determine the coefficients multiplying $D^\top$, but every pattern across buses must come from a row of D .

Lemma 1 (Congestion basis for marginal emissions). *Let $q = |\mathcal{J}|$ be the number of binding line flow constraints at d_0 . Then ℓ lies in the range of $[\mathbf{1} \ H_{\mathcal{J}}^\top]$, so there are a scalar α and a vector w of congestion weights with*

$$\ell = \alpha \mathbf{1} + H_{\mathcal{J}}^\top w, \quad r \triangleq \text{rank}[\mathbf{1} \ H_{\mathcal{J}}^\top] \leq q + 1. \quad (5)$$

Writing Φ for any matrix whose r columns are an orthonormal basis of that range and $x \in \mathbb{R}^r$ for the coordinates of ℓ in it gives $\ell = \Phi x$. We call r the congestion rank of the system.

If there are no binding line flow limits, $q = 0$ and $\ell = \alpha \mathbf{1}$, so every bus has the same LME. Each independent binding line can add at most one nodal pattern, namely its PTDF row. Binding generator limits can change α and w , since they change the dispatch response, but they add no pattern because their right sides do not depend on demand. The result parallels the Laplacian subspace characterization of admissible LMPs [Chéverez-González and DeMarco, 2009] and the energy and congestion decomposition of LMPs [He et al., 2024]. The corresponding graph view in Appendix A shows that the centered LME is harmonic away from the endpoints of binding lines [Baker and Gangmanavar, 2024].

The lemma reduces recovery of all LME values from dispatch simulations to estimating r coefficients rather than n bus values. For a demand step h at bus i , two simulations evaluate $d_0 + h e_i$ and $d_0 - h e_i$. Their central difference equals ℓ_i when both demands remain in the same critical region. Since $\ell_i = \phi_i^\top x$, each observed bus supplies one linear equation in the r unknown coefficients x .

Corollary 1 (Dispatch simulations for the LME vector). *If r rows of Φ form a nonsingular matrix, then observing ℓ_i at those r buses recovers the full vector exactly, so paired perturbations there use $2r$ dispatch simulations, or $r + 1$ with a cached base case, against $2n$ and $n + 1$ when every bus is perturbed. Fewer than r scalar observations leave x underdetermined, since they impose fewer than r independent linear constraints on it.*

The lower bound in Corollary 1 is within the declared congestion span. It does not assert that finite differences are preferable to direct derivatives. Any directional observation $\ell^\top u$ gives the equation $x^\top (\Phi^\top u)$, and r independent directions also suffice. For bus observations, pivoted QR factorization of $\Phi^\top$ selects rows that are independent and well conditioned. Thus, Corollary 1 separates the number and placement of bus perturbations. Congestion determines how many are needed, while row conditioning determines where to perturb.

3 Emissions Exceedance under Active Set Changes

Write realized demand as $d = d_0 + \xi$, where d_0 is the forecast net demand and ξ is its centered forecast error. For every direction $u \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$, assume $\mathbb{E} \exp(\lambda u^\top \xi) \leq \exp(\lambda^2 u^\top \Sigma u / 2)$, so ξ is subgaussian with variance proxy $\Sigma \succeq 0$, a matrix playing the role of a covariance that need not be diagonal. Within the nominal critical region, the emissions change is $\ell^\top \xi$, whose variance proxy is

$$v = \ell^\top \Sigma \ell = x^\top M x, \quad M = \Phi^\top \Sigma \Phi. \quad (6)$$

Although ℓ has n entries, its emissions uncertainty within the region depends only on the r coefficients x and the compressed matrix $M \in \mathbb{R}^{r \times r}$. The same congestion dimension therefore controls both recovery and uncertainty propagation while the active set remains fixed.

The nominal active constraints stay valid on the polytope $\mathcal{R} = \{\xi \mid F_j^\top \xi \leq \tau_j, j = 1, \dots, p\}$. Here $F_j \in \mathbb{R}^n$ is the outward normal of facet j and $\tau_j > 0$ is its slack at the forecast d_0 . The facets enforce inactive primal constraints and nonnegative multipliers for active inequalities (Appendix B). Applying a subgaussian tail bound to each facet and a union bound gives the region exit bound

$$\varepsilon_{\text{cr}} = \min \left\{ 1, \sum_{j=1}^p \exp \left[-\frac{\tau_j^2}{2 F_j^\top \Sigma F_j} \right] \right\}. \quad (7)$$

For Gaussian errors we replace each exponential with the exact facet probability $\Pr\{Z > \kappa_j\}$, where Z is standard normal and $\kappa_j = \tau_j / \sqrt{F_j^\top \Sigma F_j}$. We call $\kappa_{\min} = \min_j \kappa_j$ the active set margin. It is the smallest standardized distance to a critical region facet, where either an inactive primal constraint becomes binding or an active multiplier reaches zero. It is a useful summary, while the complete bound in Equation (7) retains every facet.

Theorem 1 (Operating emissions exceedance bound). *For any $z > 0$,*

$$\Pr\{\mathcal{E}(d_0 + \xi) - \mathcal{E}(d_0) \geq z\} \leq \exp[-z^2/(2v)] + \varepsilon_{\text{cr}}. \quad (8)$$

The first term bounds the carbon emission increase while the nominal active constraints hold, and Equation (7) bounds the probability of leaving that region. Keeping them separate shows which term limits the result. For a target exceedance probability δ , Equation (8) gives a finite threshold only if $\varepsilon_{\text{cr}} < \delta$; otherwise region exit alone reaches the target probability [Nilges et al., 2025]. Figure 2 shows both events.

Estimation from selected perturbations. A paired perturbation at bus i_t returns $y_t = [\mathcal{E}_{\omega_t}(d_0 + he_{i_t}) - \mathcal{E}_{\omega_t}(d_0 - he_{i_t})]/2h = \phi_{i_t}^\top x + \eta_t$, where both perturbed demands must stay in the nominal critical region and η_t is conditionally subgaussian and represents scenario sampling error in a stochastic simulator. With ridge parameter $\gamma > 0$ and $V_t = \gamma I + \sum_{s \leq t} \phi_{i_s} \phi_{i_s}^\top$, a linear regression confidence sequence under a declared $\|x\|_2 \leq \bar{x}$ bounds $\|\hat{x}_t - x\|_{V_t}$ at every step simultaneously with probability at least $1 - \delta_{\text{est}}$ [Abbasi-Yadkori et al., 2011], which permits valid stopping based on the observations. Appendix A turns this confidence set into a closed form $\bar{v}_t \geq v$. Our variance informed rule selects the next bus to maximize the reduction $\phi_{i_t}^\top V_t^{-1} M V_t^{-1} \phi_{i_t} / (1 + \phi_{i_t}^\top V_t^{-1} \phi_{i_t})$ in $\text{tr}(M V_t^{-1})$, an A -optimal experimental design weighted by M , so it covers the directions carrying emissions variance first [Fiez et al., 2019, Valko et al., 2014]. Combining $\bar{v}_t$ with Theorem 1, and assuming future forecast error is independent of simulation noise, gives the allowance $z_t = \sqrt{2\bar{v}_t \log(1/\delta_{\text{op}})}$. Its total exceedance probability is at most $\delta_{\text{op}} + \varepsilon_{\text{cr}} + \delta_{\text{est}}$ at any stopping time.

4 Numerical Results

We test 14 PGLib-OPF systems [Babaeinejadsarookolae et al., 2019] with PGLib-CO2 rates [Cho et al., 2026]. Ten have nonzero operating emissions; the other four dispatch only generators with zero benchmark rates and have a zero LME (Appendix C). A transparent DC dispatch provides exact derivatives for comparison (Appendix E).

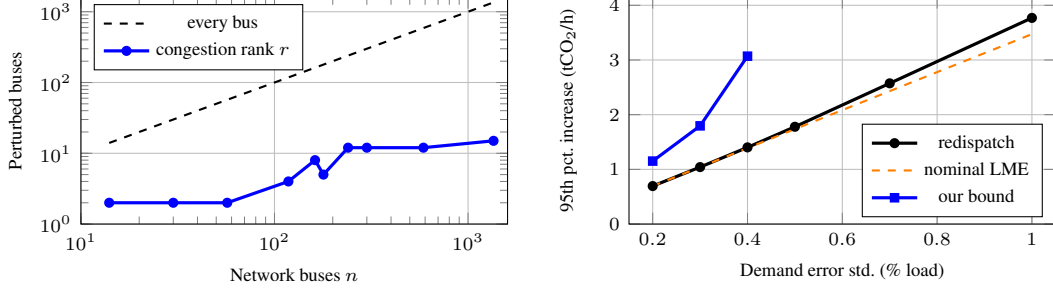


Figure 1: The congestion rank r , between 2 and 15 across ten systems, is much lower than the number of buses n (left). On the 57-bus system, the nominal LME vector underestimates the realized 95th percentile once redispatch begins, while our bound stays above it and is not reported beyond a 0.4% nodal demand forecast error standard deviation, where active set exit leaves no probability for the within region tail (right).

Simulation count. Figure 1 shows that the number of perturbed buses tracks congestion and not network size. Across 14 to 1,354 buses, the rank r ranges from 2 to 15, so the reduction n/r grows with size and reaches 90 on the largest. On the 300 bus system, 12 selected central differences recover the LME vector to 1.2×10^{-5} relative error using 24 simulations rather than 600 (Table 1). Dependent PTDF rows can make $r < q + 1$, as on the 240 bus system where $q = 13$ and $r = 12$. Omitting a binding line biases recovery by up to 44%, whereas extra lines only increase the simulation count. A residual from observations beyond the declared rank detects this error (Appendix D). Under simulation noise, the variance informed rule bounds emissions variance to within 18% after 20 simulations, before the LME vector is identified (Figure 3).

Emissions exceedance after redispatch. Figure 1 and Table 10 stress the 57-bus system with 5,000 shared Gaussian demand errors per scale. At a 1% nodal demand forecast error scale, 23.1% of draws leave the region, the nominal LME prediction underestimates the redispatch 95th percentile increase of 3.77 tonnes CO_2/h by 8.6%, and every threshold violation follows an active set change. The threshold stays available through 0.4% nodal demand forecast error, where Appendix D attributes its factor of 2.2 over the empirical quantile mostly to the subgaussian tail bound. The distributional assumption, not the facet union, causes the loosest exit bounds (Table 6).

When the bound is available. With 0.5% reserved for estimation, $\varepsilon_{\text{cr}} < 0.045$ up to nodal demand forecast error scales from 0.12% to beyond 4% of load, with no relation to network size (Table 7). Network size therefore does not indicate whether an LME remains informative under forecast uncertainty; the active set geometry must be checked at the operating point. Correlation narrows this range when $a_* > 1$ and widens it when $a_* < 1$. Here a_* compares that facet’s squared response to a common demand error with the sum of its squared responses to independent nodal errors (Appendix D). Nine of ten systems narrow, the 588 bus system has $a_* = 0.35$ and widens.

Conclusion. Small congestion rank can make recovery inexpensive on a large network, but proximity to a critical region boundary can make the recovered vector unreliable. A reported LME vector should be accompanied by its local validity. Operational use also requires models of timing, information exchange, and participant response. Future work includes sharper region exit bounds, extensions to multiperiod or nonconvex dispatch, and applying these results to other sensitivity structures such as locational marginal burden [Talkington et al., 2024].

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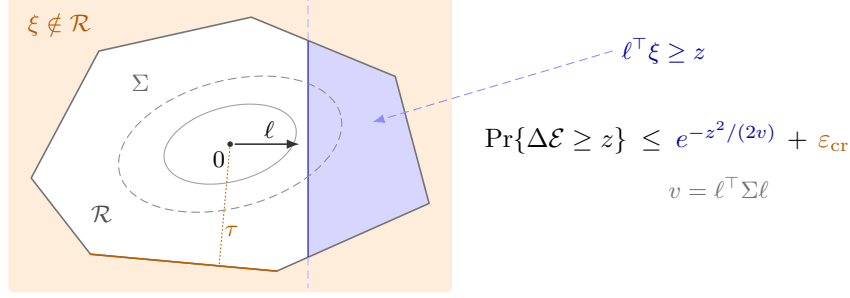


Figure 2: Geometry of Theorem 1 in forecast error space, with $\Delta\mathcal{E} = \mathcal{E}(d_0 + \xi) - \mathcal{E}(d_0)$ and level sets of Σ . Inside the critical region $\mathcal{R}$ the active set is fixed and $\Delta\mathcal{E} = \ell^\top \xi$, so reaching z means entering the blue set, the halfspace inside $\mathcal{R}$. Every facet contributes to ε_{cr} , dominated by the facet closest to the nominal operating point, at slack τ .

A Proofs

A.1 Congestion Basis for Marginal Emissions

For a fixed active set, stationarity and feasibility are

$$Qg + a + K^\top \nu = 0, \quad Kg = b + Dd.$$

Here ν collects the multipliers for the active equalities. Eliminating ν gives Equation (3), and

$$\ell = D^\top (KQ^{-1}K^\top)^{-1} KQ^{-1}c.$$

The power balance equality contributes **1**. An active upper or lower line constraint contributes a signed row of $H_{\mathcal{J}}$. Generator bound rows contribute zero because their right sides do not depend on demand. This proves Equation (5).

Graph interpretation. Let A denote the branch bus incidence matrix, B the diagonal matrix of branch susceptances, and $L = A^\top B A$ the weighted network Laplacian of a connected network. In a shuntless network without phase shifters, the DC power flow relation gives $H = B A L^\dagger$, where $L^\dagger$ is the pseudoinverse of L [Chéverez-González and DeMarco, 2009, Baker and Gangammanavar, 2024]. Substituting this identity into Equation (5) gives

$$\ell - \alpha \mathbf{1} = L^\dagger A^\top B_{\mathcal{J}} w \implies L(\ell - \alpha \mathbf{1}) = A_{\mathcal{J}}^\top B_{\mathcal{J}} w,$$

because the source on the right sums to zero. This is a discrete Poisson equation whose source is supported at no more than $2q$ endpoints of the binding lines. We use *harmonic* in the graph theoretic sense. A bus value vector z is harmonic at bus i when $(Lz)_i = 0$, which means that z_i equals the susceptance weighted average of its neighboring values [Chung, 1997, Shuman et al., 2013].

Let $\mathcal{T}$ collect the binding line endpoints. When $q > 0$, every component remaining after $\mathcal{T}$ is removed is adjacent to $\mathcal{T}$. If two vectors in the congestion span agree on $\mathcal{T}$, their difference is zero there and harmonic at every other bus. Uniqueness of the discrete Dirichlet problem therefore makes the difference zero everywhere [Chung, 1997]. Thus, the values on $\mathcal{T}$ determine the full vector. If $q = 0$, the vector is uniform and one bus value suffices. Independent row selection reduces these boundary observations to the minimum r .

A.2 Recovery from Selected Buses

Let I contain r buses such that Φ_I is nonsingular. Their observations satisfy $\ell_I = \Phi_I x$, hence $x = \Phi_I^{-1} \ell_I$ and $\ell = \Phi \Phi_I^{-1} \ell_I$. This proves Corollary 1. Without the congestion structure, fewer than n sparse observations cannot distinguish a zero vector from a marginal emissions vector supported at an unobserved bus.

A.3 Exceedance Bound under Active Set Changes

On the event $\{\xi \in \mathcal{R}\}$ the dispatch stays affine because the active constraints are fixed, so $\mathcal{E}(d_0 + \xi) - \mathcal{E}(d_0) = \ell^\top \xi$ and

$$\Pr\{\mathcal{E}(d_0 + \xi) - \mathcal{E}(d_0) \geq z\} \leq \Pr\{\ell^\top \xi \geq z\} + \Pr\{\xi \notin \mathcal{R}\}.$$

The Chernoff bound gives the first term in Equation (8). Applying the same upper tail bound to every event $\{F_j^\top \xi > \tau_j\}$ and then taking a union bound gives Equation (7). This proves Theorem 1.

A.4 Confidence Bound for Emissions Variance

Let $e = x - \hat{x}_t$. Expanding the variance proxy gives

$$x^\top Mx = \hat{x}_t^\top M\hat{x}_t + 2e^\top M\hat{x}_t + e^\top Me.$$

The confidence event implies $\|V_t^{1/2}e\|_2 \leq \beta_t$. The Cauchy–Schwarz inequality bounds the cross term and a Rayleigh quotient bounds the quadratic term, so

$$\bar{v}_t = \hat{x}_t^\top M\hat{x}_t + 2\beta_t\|V_t^{-1/2}M\hat{x}_t\|_2 + \beta_t^2\lambda_{\max}(V_t^{-1/2}MV_t^{-1/2}) \geq v. \quad (9)$$

The confidence event holds simultaneously at every step, so the result also holds at a stopping time selected from the data.

B Critical Region Construction

Write an inactive inequality as $A_j g \leq b_j + E_j d$. With $g = g_0 + G\xi$, primal feasibility becomes

$$(A_j G - E_j)\xi \leq b_j + E_j d_0 - A_j g_0.$$

Thus this constraint contributes facet normal $F_j^\top = A_j G - E_j$ and slack $\tau_j = b_j + E_j d_0 - A_j g_0$. For an active inequality, let $\mu_i(\xi) = \mu_{i,0} + N_i \xi$ denote its KKT multiplier. Dual feasibility becomes

$$(-N_i)\xi \leq \mu_{i,0}, \quad N = -(KQ^{-1}K^\top)^{-1}D.$$

It contributes facet normal $F_j^\top = -N_i$ and slack $\tau_j = \mu_{i,0}$. The regularity assumption makes every retained slack positive at d_0 . The implementation removes zero normals and dependent active rows, and it converts demand directions from per unit to MW before evaluating Equation (7).

C Full Screening Results

Table 1: Congestion sets the rank on every retained system, and the reduction n/r improves with system size. Here q counts binding line constraints, r is the congestion basis rank, E_{span} is the span residual, and E_P is the graph Poisson residual.

System	n	q	r	n/r	E_{span}	E_P
14	14	1	2	7.0	1.4e-15	3.5e-14
30	30	1	2	15.0	3.1e-15	8.1e-14
57	57	1	2	28.5	4.3e-14	2.7e-13
118	118	3	4	29.5	4.9e-15	2.9e-14
162	162	7	8	20.2	8.6e-15	1.3e-14
179	179	4	5	35.8	9.6e-15	3.6e-14
240	240	13	12	20.0	2.7e-14	1.6e-13
300	300	11	12	25.0	1.1e-13	3.6e-13
588	588	11	12	49.0	1.1e-13	2.0e-11
1354	1354	14	15	90.3	2.1e-14	2.6e-14

Table 2: Four screened systems carry no operating emissions under the direct emission boundary, so their LME vector is zero and the study does not apply.

System	exclusion reason
73	zero operating emissions under direct factors
197	zero operating emissions under direct factors
500	zero operating emissions under direct factors
793	zero operating emissions under direct factors

At the selected operating point, the four systems of Table 2 dispatch only generators that the PGLib-CO2 lookup assigns a zero direct operating rate. All four also miss our declared sensitivity tolerance of 10^{-8} , with relative disagreements between the analytical sensitivity and the numerical active set derivative running from 8×10^{-8} to 0.42, and three of the four have an optimality system with condition number above 10^9 . The exception is the 197 bus system, whose optimality system is well conditioned but whose disagreement is the largest of the four. We read both patterns as degeneracy of the active set at those operating points, and we exclude the systems on the emissions criterion, which can be checked before computing any sensitivity.

D Additional Numerical Details

This appendix collects the supporting studies for Section 4, namely the perturbation step sweep in Table 3, the misdeclared binding set in Table 4, the congestion sweep in Table 5, the tightness of the exit bound in Table 6, the region geometry across systems in Table 7, the correlated forecast error in Figure 4 and Table 8, the ridge sensitivity in Table 9, and the design comparison in Figure 3.

Write the correlated proxy as $\Sigma_\rho = \rho \sigma \sigma^\top + (1 - \rho) \text{diag}(\sigma^2)$, which leaves every marginal standard deviation at σ_i . Then $F_j^\top \Sigma_\rho F_j = (1 - \rho + \rho a_j) \sum_i (\sigma_i F_{ji})^2$ with $a_j = (\sigma^\top F_j)^2 / \sum_i (\sigma_i F_{ji})^2$, so the standardized distance obeys $\kappa_j(\rho) = \kappa_j(0) / \sqrt{1 - \rho + \rho a_j}$ exactly. Cauchy Schwarz gives $a_j \in [0, n]$, with $a_j = n$ when the facet responds equally to every bus and $a_j = 0$ when it is a pure differential, and correlation costs margin whenever $a_j > 1$. Table 7 reports the ratio at the nearest facet of each system, where the measured $\kappa_\star(0.5)$ matches this prediction to every digit shown.

The operating point search tests load scales 1.0, 1.1, 1.2 and line limit scales 1.0, 0.9, $\dots$, 0.3 in fixed order. We add 10^{-4} to every quadratic cost coefficient so that the local derivative is unique. The base uncertainty proxy for every system draws independent Gaussian nodal demand forecast errors with standard deviation equal to one percent of nominal demand and a 0.25 MW floor. The stress test in Section 4 multiplies that proxy by scales from 0.2 to 1.0, so the reported demand error standard deviation of 0.2% to 1.0% of load is the product of the two.

Table 3: The perturbation step must be small enough to keep both perturbed demands inside the critical region. A 1 MW step recovers the vector to 1.2×10^{-5} , 10 MW leaves the region and costs 4.2%, and 100 MW makes the dispatch infeasible. Facet use is the largest fraction of a facet slack consumed by any selected perturbation, and a value above one predicts the loss of accuracy.

step h (MW)	inside region	facet use	recovery error
0.1	yes	0.08	2.2e-04
1.0	yes	0.77	1.2e-05
10.0	no	7.75	4.2e-02
50.0	no	38.73	3.0e-01
100.0	no	77.46	infeasible
250.0	no	193.65	infeasible
500.0	no	387.30	infeasible

Table 4: A declared binding set that omits a binding line biases recovery, while one that names extra lines costs simulations but stays exact. Errors are relative and taken over all single line omissions on the 300 bus system.

declared binding set	variants	rank r	median error	worst error
exact	1	12	1.2e-13	1.2e-13
omit one binding line	11	11	3.4e-02	4.4e-01
add 2 slack lines	1	14	1.1e-13	1.1e-13
add 5 slack lines	1	17	1.0e-13	1.0e-13

Table 5: Congestion sweep on the 300 bus system. The number of binding lines and the rank both depend on the operating point.

line limit multiplier	binding lines q	rank r	n/r
1.3	5	6	50.0
1.2	6	7	42.9
1.1	8	9	33.3
1.0	11	12	25.0
0.9	10	11	27.3

Table 6: The Gaussian column sits within Monte Carlo error of the empirical frequency on every tested system, while the distribution free subgaussian column overshoots by up to a factor of five. Concentration is the share of the Gaussian bound carried by the single nearest facet, and it exceeds 0.93 on eight of ten systems. Frequencies use 2,000 Gaussian draws.

System	facets	$\kappa_{\min}$	sub-G.	Gauss.	exit freq.	conc.
14	46	7.86	0.000	0.000	0.000	1.00
30	89	26.65	0.000	0.000	0.000	1.00
57	168	0.70	0.785	0.243	0.236	1.00
118	422	1.55	0.436	0.083	0.080	0.73
162	581	3.12	0.008	0.001	0.000	1.00
179	556	1.12	0.596	0.141	0.143	0.94
240	1034	1.44	0.908	0.176	0.166	0.43
300	891	0.79	0.731	0.215	0.222	1.00
588	1468	0.21	1.000	0.420	–	0.99
1354	4239	2.36	0.062	0.009	–	1.00

Table 7: The bound is available across the size range, and the aggregate response ratio a_* of the nearest facet predicts the sign of the correlation effect exactly. Availability is the largest nodal demand forecast error standard deviation, as a percentage of load, at which the exit term still leaves room in the exceedance allowance. Systems with $a_* > 1$ lose range under correlation and the 588 bus system, with $a_* = 0.35$, gains it. The facet type does not predict the sign.

System	$\kappa_{\min}$	a_*	nearest facet	avail. $\rho=0$	avail. $\rho=0.5$
14	7.86	5.9	generator	>4.0	2.50
30	26.65	6.1	generator	>4.0	>4.0
57	0.70	5.6	generator	0.41	0.23
118	1.55	35.6	generator	0.86	0.21
162	3.12	26.3	line	1.84	0.50
179	1.12	17.1	line	0.66	0.22
240	1.44	1.1	generator	0.72	0.41
300	0.79	24.5	generator	0.47	0.13
588	0.21	0.3	generator	0.12	0.15
1354	2.36	61.6	generator	1.39	0.25

Table 8: Correlated demand error narrows the usable range on the 57 bus system. The demand error standard deviation at each bus is held fixed, so the only change is the correlation ρ between buses. Thresholds are in tonnes CO2/h and a dash means the exit term has consumed the exceedance allowance.

ρ	demand error std. (% load)	$\kappa_{\min}$	exit freq.	ε_{cr}	threshold
0.0	0.20	3.48	0.000	0.000	1.15
0.0	0.30	2.32	0.007	0.010	1.80
0.0	0.40	1.74	0.037	0.041	3.07
0.0	0.50	1.39	0.073	0.082	—
0.0	0.70	0.99	0.152	0.160	—
0.0	1.00	0.70	0.231	0.243	—
0.3	0.20	2.25	0.010	0.012	2.17
0.3	0.30	1.50	0.064	0.066	—
0.3	0.40	1.13	0.122	0.130	—
0.3	0.50	0.90	0.173	0.184	—
0.3	0.70	0.64	0.250	0.260	—
0.3	1.00	0.45	0.317	0.326	—
0.5	0.20	1.91	0.025	0.028	2.86
0.5	0.30	1.28	0.100	0.101	—
0.5	0.40	0.96	0.167	0.169	—
0.5	0.50	0.77	0.220	0.222	—
0.5	0.70	0.55	0.286	0.292	—
0.5	1.00	0.38	0.346	0.351	—
0.8	0.20	1.61	0.053	0.054	—
0.8	0.30	1.07	0.144	0.142	—
0.8	0.40	0.80	0.209	0.211	—
0.8	0.50	0.64	0.261	0.260	—
0.8	0.70	0.46	0.323	0.323	—
0.8	1.00	0.32	0.374	0.374	—

Table 9: The confidence radius grows like $R\sqrt{r \log(1/\gamma)}$ through the log determinant term and like $\sqrt{\gamma} \bar{x}$ through the declared norm bound, which places an interior optimum at the value used in the paper. Values are medians over 100 trials after 40 dispatch simulations on the 57 bus system.

γ	$\bar{v}_t/v$	threshold inflation
10^{-14}	1.155	1.075
10^{-12}	1.144	1.070
10^{-10}	1.134	1.065
10^{-8}	1.125	1.061
10^{-6}	1.143	1.069
10^{-4}	1.472	1.213
10^{-2}	7.369	2.715
10^{-1}	32.603	5.710
10^0	105.577	10.275

The recovery study on the 300 bus system uses a 1 MW central difference, and a pivoted QR factorization selects the perturbation buses. The noisy study uses

$$\beta_t = R \sqrt{\log \frac{\det V_t}{\det(\gamma I)} + 2 \log(1/\delta_{\text{est}}) + \sqrt{\gamma} \bar{x}}$$

with $R = 0.02$ tonnes CO2/MWh, which is 2.9% of the mean marginal emission rate on that system and stands for the scenario sampling error a stochastic simulator would carry, together with $\gamma = 10^{-8}$, $\delta_{\text{est}} = 0.005$, and $\bar{x} = 5\sqrt{n}$. The realized $\|x\|_2 = 12.5$ sits well below $\bar{x} = 86.6$, and since $\sqrt{\gamma} \bar{x} \approx 10^{-3}$ here, the declared bound contributes almost nothing to β_t and a deployment would

only need it to be finite. One noisy observation represents a pair of load perturbations and therefore requires two dispatch simulations when the base case is not cached. After 40 simulations, uniform bus selection has median $\sqrt{\bar{v}_t/v} = 433$, because it leaves directions relevant to M nearly unobserved. It is an ill conditioned diagnostic rather than evidence of a comparable advantage over leverage selection.

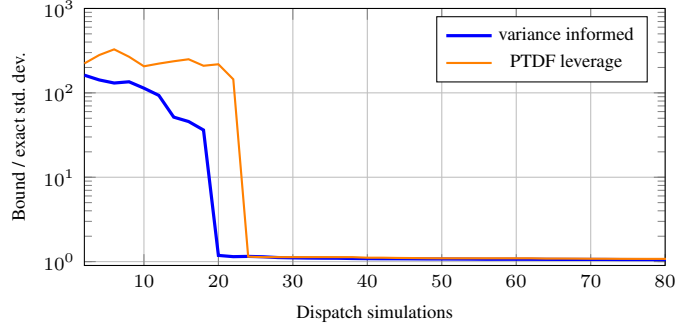


Figure 3: Weighting the design by M bounds the emissions variance before the LME vector is identified. On the 300 bus system, where $r = 12$, variance informed selection reaches a usable bound after 20 dispatch simulations, while leverage selection must cover the full basis first and overestimates the standard deviation by a factor of 218 at that point. Once both cover the basis the advantage shrinks to 2.6%, with interquantile ranges of $[1.079, 1.091]$ and $[1.102, 1.122]$ that do not overlap.

The separate region diagnostic reuses 50,000 standard Gaussian draws across 18 forecast error scales on the 118 bus system. The complete experiment on the 57 bus system uses 5,000 draws at each scale and the Gaussian facet union bound. Summary exit estimates use 2,000 draws on systems with up to 300 buses, and we do not report an empirical frequency for larger systems.

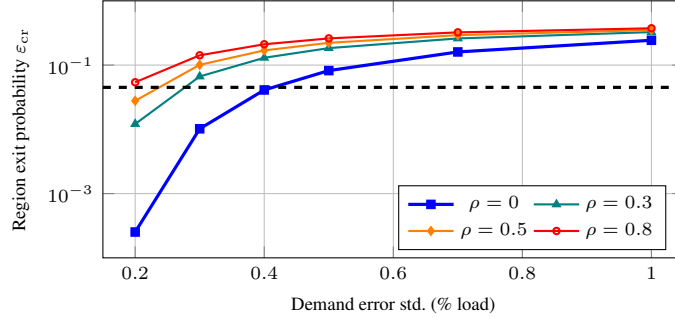


Figure 4: On the 57-bus system, the nearest facet has aggregate response ratio $a_* = 5.6$, so correlation costs margin and the exit probability crosses the remaining exceedance allowance, dashed, at a smaller demand error scale as ρ grows. The crossing moves from 0.41% at $\rho = 0$ to 0.23% at $\rho = 0.5$. Table 7 reports a_* for every system, including the one where it is below one and the ordering reverses.

Table 10: Complete emissions experiment on the 57 bus system. Thresholds are in tonnes CO2/h. The selected perturbation column is the median over 100 trials of variance informed selection after 40 dispatch simulations, and it assigns 0.5% allowed exceedance probability to estimation. A dash means the remaining 5% exceedance allowance leaves no positive probability for the within region tail. The solve count includes every realization outside the region and 100 validation draws inside it.

demand forecast error std. (% load)	$\kappa_{\min}$	exit freq.	ε_{cr}	empirical $q_{.95}$	known v	$z_{.95}$	selected perturb. $z_{.95}$	solves
0.20	3.48	0.000	0.000	0.69		1.07	1.15	100
0.30	2.32	0.007	0.010	1.04		1.60	1.80	136
0.40	1.74	0.037	0.041	1.40		2.13	3.07	283
0.50	1.39	0.073	0.082	1.78		2.67	–	466
0.70	0.99	0.152	0.160	2.57		3.73	–	860
1.00	0.70	0.231	0.243	3.77		5.33	–	1255

The reported threshold at a nodal demand forecast error standard deviation of 0.4% exceeds the empirical 95th percentile by a factor of 2.2, which decomposes into 1.49 for the subgaussian tail bound, 1.25 for splitting the exceedance allowance with the exit term, and 1.15 for estimation after 40 simulations. Under a Gaussian forecast error, the exact quantile replaces the Chernoff step of Theorem 1, which isolates the price of the weaker subgaussian assumption. With a known v , that refinement gives 1.43 tonnes CO2/h against an empirical 1.40, and it stays within 3% at every scale where the answer is available. At 0.7% and above, where the answer is not available, the same Gaussian threshold falls below the true quantile, which is the bias that the exit term prevents.

E Reproducibility

The Julia implementation and numerical results are available in an anonymized repository at <https://anonymous.4open.science/r/LMEs-and-Carbon-Concentration-6372>. The repository includes experiment scripts, benchmark data and provenance, generated CSV outputs, and automated tests. The experiment manifest records package versions, random seeds, solver settings, exclusion criteria, uncertainty parameters, and sample counts. The README provides instructions for running a short validation case, the test suite, and the complete numerical pipeline.